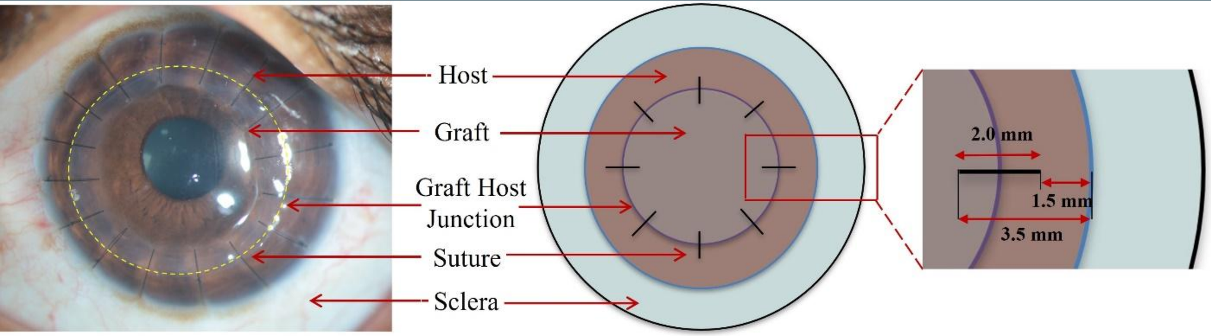
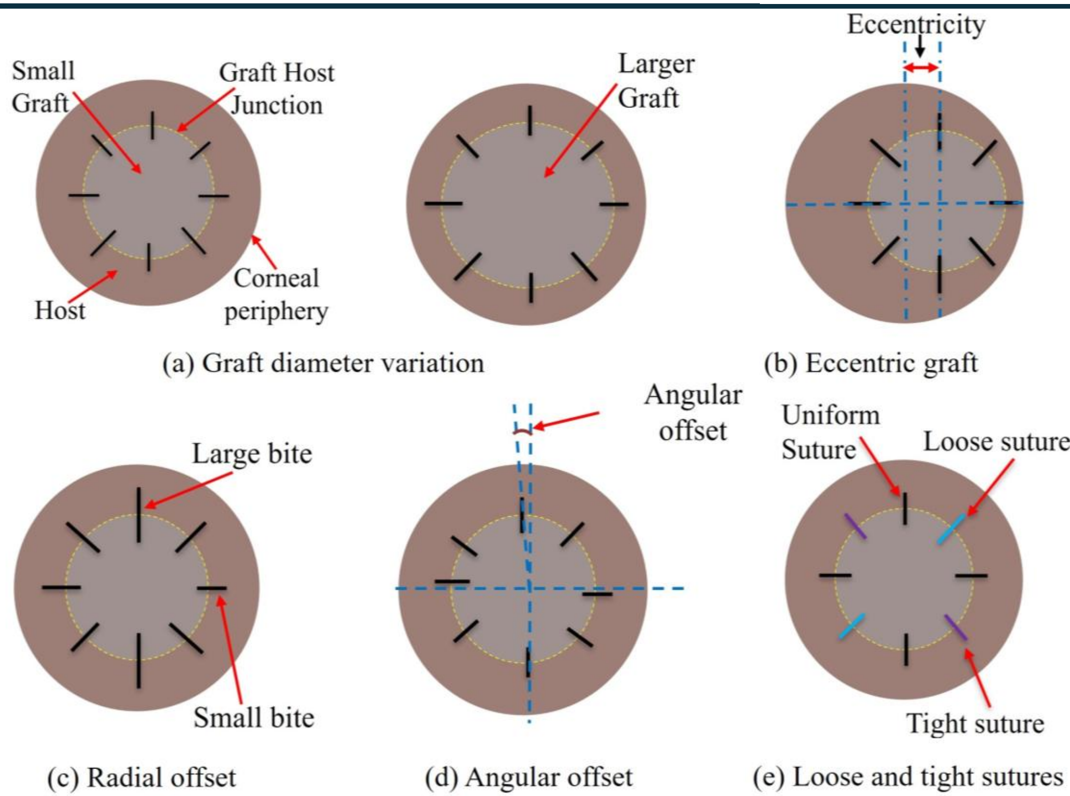
भारतीय प्रौद्योगिकी संस्थान हैदराबाद
Indian Institute of Technology Hyderabad

PK FEM Model

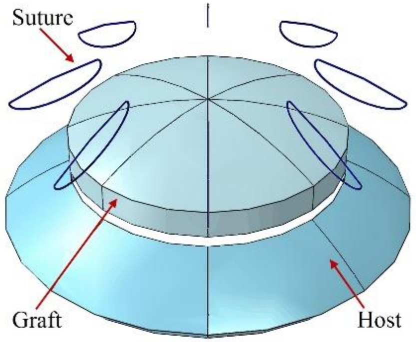
LVPEI
So that all may see



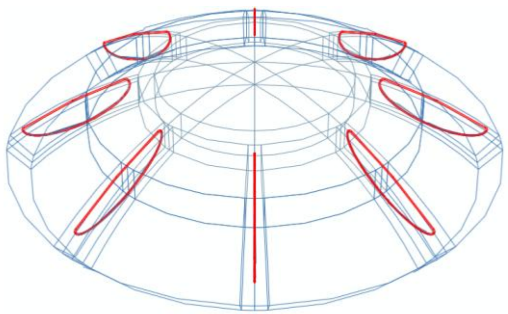
Post PK cornea with the geometrical representation of sutures



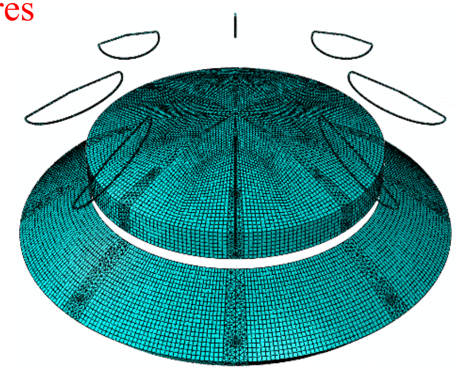
Schematic diagram for graft for different surgical scenarios



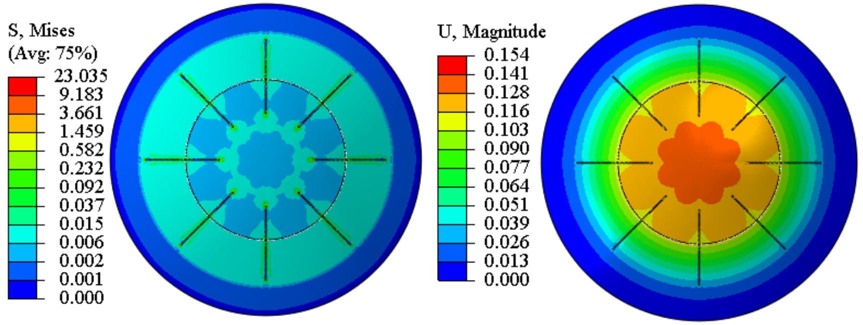
PK Model with 8 solid sutures



Wireframe view



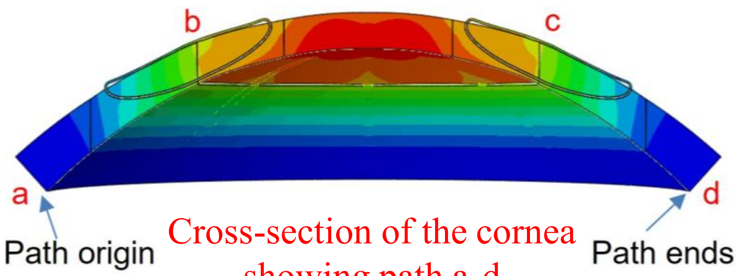
Meshed model



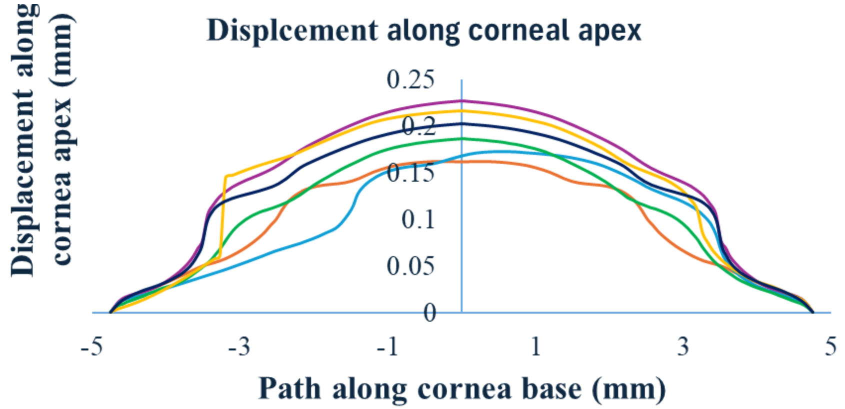
Stress and Displacement in Graft diameter 5mm

- Larger grafts and eccentric placement increase junction stress and apex shift.
- Loose-tight sutures cause high stress, risking damage.
- Angular/radial offsets lead to uneven stress and asymmetric deformation.
- Eccentric graft displaces the apex position.
- The sixteen-suture model ensures uniform stress and stable shape.

PK FEM Optomechanical Analysis

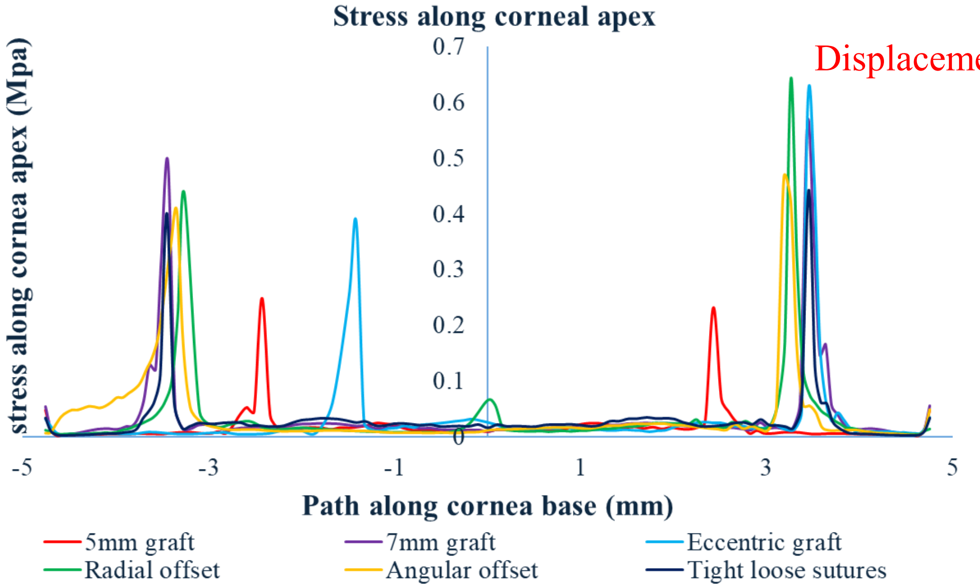


Cross-section of the cornea showing path a-d



Displacement distribution for the solid sutures model

- Loose-tight: Max stress, risk of graft failure.
- Angular/Eccentric: High displacement, optical distortion.
- Radial offset: Uneven tension, high stress, and error.
- Small graft: Low deformation, steep curvature.
- Sixteen sutures: Best stability and optical balance.



Stress distribution for solid sutures model

Solid Sutures Disorientation Analysis

Disorientation scenarios	Max Suture Tension (mN)	Max apex displacement (mm)	Max stress (MPa)	Mean refractive power (D)
Small graft	12.20	0.158	17.26	65.42
Large graft	13.50	0.194	19.15	60.65
Radial offset	19.57	0.175	27.67	64.01
Angular offset	20.50	0.216	32.38	62.58
Eccentric graft	26.83	0.173	37.94	63.70
Loose-tight suture	39.40	0.203	207.30	60.62
Sixteen sutures model	8.08	0.180	11.425	61.01

Solid sutures Table showing mechanical and optical analysis

- Solid sutures cause higher stress but smoother displacement.